DSC 206: Homework 3

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Assignment 1 *Consider the two points $\boldsymbol{x} = (1, 0, 1, 0, 1, 0)$ and $\boldsymbol{y} = (0, 1, 1, 0, 1, 0)$. Compute the distance between them under the L_0 (Hamming), L_1, L_2 (Euclidean), and L_∞ distances.*

Solution. The coordinate-wise differences are $\boldsymbol{x} - \boldsymbol{y} = (1, -1, 0, 0, 0, 0)$.

$$\begin{aligned}
L_0(\boldsymbol{x}, \boldsymbol{y}) &= \#\{i : x_i \neq y_i\} = 2, \\
L_1(\boldsymbol{x}, \boldsymbol{y}) &= \sum_i |x_i - y_i| = 1 + 1 = 2, \\
L_2(\boldsymbol{x}, \boldsymbol{y}) &= \sqrt{\sum_i (x_i - y_i)^2} = \sqrt{2}, \\
L_\infty(\boldsymbol{x}, \boldsymbol{y}) &= \max_i |x_i - y_i| = 1.
\end{aligned}$$

Assignment 2 *On the space of nonnegative integers, decide whether each of the following is a distance measure. Recall the axioms: (i) $d(x, y) \geq 0$, (ii) $d(x, y) = 0 \iff x = y$, (iii) $d(x, y) = d(y, x)$, (iv) triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$.*

(a) $\max(x, y)$ *— the larger of x and y;*

(b) $\operatorname{diff}(x, y) = |x - y|$*;*

(c) $\operatorname{sum}(x, y) = x + y$*.*

Solution. **(a)** $\max(x, y)$ **is *not* a distance.** It violates axiom (ii): for any $x > 0$, $\max(x, x) = x \neq 0$, so $d(x, x) \neq 0$.

(b) $\operatorname{diff}(x, y) = |x - y|$ **is a distance.**

(i) $|x - y| \geq 0$ by definition of absolute value.

(ii) $|x - y| = 0 \iff x - y = 0 \iff x = y$.

(iii) $|x - y| = |-(y - x)| = |y - x|$.

(iv) $|x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z|$ by the triangle inequality for $|\cdot|$ on $\mathbb{R}$.

(c) $\mathrm{sum}(x, y) = x + y$ **is *not* a distance.** It violates axiom (ii): $d(2, 2) = 4 \neq 0$.

Assignment 3 *Starting from the family of minHash functions (so a single hash matches with probability equal to the Jaccard similarity s), describe the effect on the matching probability of the following constructions. Write the probability as a function of s.*

(a) 2*-way AND followed by* 3*-way OR.*
(b) 3*-way OR followed by* 2*-way AND.*
(c) 2*-way AND, then* 2*-way OR, then* 2*-way AND.*
(d) 2*-way OR, then* 2*-way AND, then* 2*-way OR, then* 2*-way AND.*

Solution. Recall that an r-way AND on a family with collision probability p produces collision probability p^r, and an r-way OR produces $1 - (1 - p)^r$. We compose stage by stage starting from $p_0 = s$.

(a) AND-2 gives s^2; OR-3 gives

$$p(s) = 1 - (1 - s^2)^3.$$

(b) OR-3 gives $1 - (1 - s)^3$; AND-2 gives

$$p(s) = \big(1 - (1 - s)^3\big)^2.$$

(c) AND-2 gives s^2; OR-2 gives $1 - (1 - s^2)^2$; AND-2 gives

$$p(s) = \big(1 - (1 - s^2)^2\big)^2.$$

(d) OR-2 gives $1 - (1 - s)^2 = 2s - s^2$; AND-2 gives $(2s - s^2)^2 = s^2(2 - s)^2$; OR-2 gives $1 - (1 - s^2(2 - s)^2)^2$; AND-2 gives

$$p(s) = \Big(1 - \big(1 - s^2(2 - s)^2\big)^2\Big)^2.$$

Assignment 4 *Consider a Bloom filter with $n = 8$ bits of memory, a set S of $m = 100$ allowed keys, and $k = 3$ independent random hash functions. Compute (i) the probability that an element of S is blocked, and (ii) the probability that an element not in S is going to pass the filter (i.e. not be blocked by the filter). Do not use the simplified large-n approximation.*

Solution. "Pass" means the filter accepts (all k probed bits equal 1); "block" means it rejects (at least one probed bit equals 0).

(i) Element in S. A Bloom filter has *no false negatives*: every bit set during the insertion of $x \in S$ remains set, so on a query for x all k probed bits are 1 and the filter accepts. Hence

$$\Pr[\text{block} \mid x \in S] = 0.$$

(ii) Element not in S. During insertion, each of the k hash functions sends each of the m keys to a uniformly random bit, for a total of $km = 300$ independent uniform throws into $n = 8$ bins. A particular bit stays 0 iff none of those throws lands on it, so

$$\Pr[\text{bit } i = 0] = \left(1 - \tfrac{1}{n}\right)^{km} = \left(\tfrac{7}{8}\right)^{300}, \qquad \Pr[\text{bit } i = 1] = 1 - \left(\tfrac{7}{8}\right)^{300}.$$

A query on $x \notin S$ probes $k = 3$ uniformly random bits, and the filter passes iff all of them are 1. Treating the k probed bits as independent (this is the standard textbook assumption — it ignores both that the query hashes may coincide and the weak negative correlation between bit states),

$$\boxed{\Pr[\text{pass} \mid x \notin S] = \left(1 - \left(1 - \tfrac{1}{n}\right)^{km}\right)^{k} = \left(1 - \left(\tfrac{7}{8}\right)^{300}\right)^{3}.}$$

The problem instructs us not to apply the simplified large-n approximation $(1 - 1/n)^{km} \approx e^{-km/n}$; we kept the exact $(7/8)^{300}$ instead. Numerically,

$$\left(\tfrac{7}{8}\right)^{300} = e^{300 \ln(7/8)} \approx e^{-40.06} \approx 4.4 \times 10^{-18},$$

so

$$\Pr[\text{pass} \mid x \notin S] \approx \left(1 - 4.4 \times 10^{-18}\right)^{3} \approx 1 - 1.3 \times 10^{-17} \;\approx\; 1.$$

With only $n = 8$ bits storing $m = 100$ keys via $k = 3$ hashes, the load factor $km/n = 37.5$ is so high that every bit is essentially certain to be 1, and the filter accepts essentially every non-member.

Assignment 5 *Consider the data stream*

$$1,\ 2,\ 6,\ 3,\ 2,\ 2,\ 3,\ 1,\ 6,\ 6.$$

(a) What is the number of distinct items?

(b) Run the Flajolet–Martin algorithm with alphabet size $m = 6$ and hash $h(x) = 3x + 1 \bmod 7$. Show your work.

(c) Find the frequency of each element.

Solution. **(a) Distinct count.** The distinct values appearing in the stream are $\{1, 2, 3, 6\}$, so the true number of distinct items is 4.

(b) Flajolet–Martin. Apply the hash to each distinct stream value. Since h takes values in $\{0, 1, \ldots, 6\}$, we represent each value with $\lceil \log_2 7 \rceil = 3$ bits and let $r(x)$ denote the number of trailing zeros in the binary expansion of $h(x)$.

x	$h(x) = 3x + 1 \bmod 7$	binary (3 bits)	$r(h(x))$
1	4	100	2
2	0	000	3
3	3	011	0
6	5	101	0

We use the convention that $h(x) = 0$ contributes the maximum tail-length $\lceil \log_2 7 \rceil = 3$. The algorithm tracks $R = \max_x r(h(x))$ as the stream is processed; tracing through $1, 2, 6, 3, 2, 2, 3, 1, 6, 6$ the running maximum becomes $2, 3, 3, 3, 3, 3, 3, 3, 3, 3$. Hence $R = 3$, and the FM estimate is

$$\widehat{D} = 2^R = 2^3 = 8.$$

This overestimates the true value 4, which is typical for a single FM register on a tiny stream — in practice one averages many independent hash functions to reduce variance. (If one instead drops the $h = 0$ value as ill-defined, then $R = 2$ and $\widehat{D} = 4$, matching the truth exactly.)

(c) Frequencies. The stream has length 10. Counting occurrences,

$$f(1) = \tfrac{2}{10} = 0.2, \quad f(2) = \tfrac{3}{10} = 0.3, \quad f(3) = \tfrac{2}{10} = 0.2, \quad f(6) = \tfrac{3}{10} = 0.3.$$